Evaluating Consequential Meaning Errors in Hindi Citizen Grievances: A Challenge Set Analysis of COMETKiwi

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***Abstract*—Automatic translation evaluation is increasingly relevant to multilingual public-service communication, where small meaning changes can affect how a citizen's request is interpreted. This paper presents a controlled challenge-set analysis of COMETKiwi on synthetic English–Hindi citizen-grievance translation pairs. The set contains 100 pairs across ten intended administrative meaning-error categories, with one intended faithful Hindi candidate and one deliberately corrupted candidate per pair. We evaluate whether the metric assigns a higher quality-estimation score to the faithful candidate. Across 100 pairs, COMETKiwi ranked the faithful candidate higher in 95 cases and the corrupted candidate higher in five, yielding 95% strict pairwise agreement. The mean score difference was 0.041884, and the median was 0.016785. Time-reference errors had the lowest category-level agreement, with eight of ten pairs ranked correctly. The exploratory analysis also examines candidate-score distributions and pairwise score margins. These findings characterize metric behavior on a small, synthetic, category-balanced diagnostic set; they do not establish performance on naturally occurring grievances or demonstrate real-world harm. The pair labels reflect intended synthetic constructions and have not been independently validated by bilingual human annotators.**

***Index Terms*—machine translation evaluation, quality estimation, Hindi, citizen grievances, COMETKiwi, challenge set, semantic fidelity**

I. Introduction

Public-service systems increasingly receive requests in multiple languages. In this setting, translation errors involving dates, amounts, locations, urgency, or requested actions may change how a request is understood. General translation-quality scores can summarize overall quality, but aggregate scores may obscure whether an evaluator distinguishes a faithful translation from a minimally altered candidate.

This study examines the pairwise ranking behavior of COMETKiwi, a reference-free machine-translation quality-estimation metric, on a controlled English–Hindi challenge set designed around administrative meaning changes. Each item contains an English source, an intended faithful Hindi candidate, and a corrupted Hindi candidate intended to alter an administrative detail. The experiment asks whether the metric assigns a higher score to the faithful candidate.

The contributions are a synthetic, category-balanced set of 100 candidate pairs, a pairwise and category-level score-margin analysis, an exploratory analysis of candidate scores, and an examination of five ranking inversions and low-margin comparisons. This is a diagnostic analysis rather than a new evaluation metric or a claim about deployment readiness. Because the set is synthetic and its semantic labels have not been independently validated by bilingual annotators, the findings should be interpreted as a pilot characterization of metric behavior.

II. Research Questions

This study addresses the following questions:

- **RQ1:** How often does COMETKiwi rank the intended faithful Hindi candidate above the corrupted candidate?
- **RQ2:** How do pairwise ranking outcomes and score margins vary across the ten intended error categories?
- **RQ3:** What characteristics are present in cases where the corrupted candidate receives a higher score?
- **RQ4:** What do the candidate-score distributions and pairwise margins reveal about metric behavior on this constructed dataset?

III. Related Work

Automatic machine-translation evaluation has moved beyond lexical overlap toward learned metrics that model semantic adequacy and translation quality. COMETKiwi provides reference-free quality estimation and was introduced in the context of the WMT quality-estimation shared task [1]. Reference-free evaluation is useful when a gold reference translation is unavailable, but its scores remain estimates rather than guarantees of semantic correctness.

More recent work, including xCOMET, explores finer-grained evaluation and error-span explanations [2]. Such work motivates evaluating not only average scores but also behavior on targeted error cases. For multilingual Indic settings, systems such as IndicTrans2 illustrate ongoing efforts to improve translation across Indian languages [3]. The present work is complementary: it does not compare translation systems, but probes whether a quality-estimation metric distinguishes intended faithful and corrupted Hindi candidates in a constrained administrative domain.

The study makes a narrow contribution: a controlled Hindi citizen-grievance challenge-set analysis. It does not claim to

be the first study of translation metrics, Hindi translation, or administrative translation errors.

IV. Exploratory Data Analysis

A. Dataset Composition

The supplied result files contain 200 candidate-level scores and 100 paired comparisons. Each pair contains one intended faithful Hindi candidate and one corrupted candidate. The challenge set is balanced by construction: each of the ten intended error categories contains ten pairs, giving 20 candidate translations per category.

The categories are negation and service status, date, duration, amount, location, location detail, requested action, department-routing correctness, time reference, and urgency. Figure 1 illustrates the number of pairs in each category.

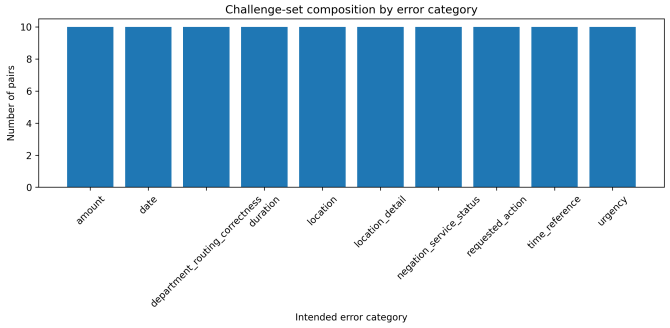


Fig. 1. Number of paired examples in each intended error category. Each category contains ten pairs.

B. Candidate Score Distributions

Across the 100 intended faithful candidates, the mean COMETKiwi score was approximately 0.8557, with a standard deviation of 0.0348. Across the 100 corrupted candidates, the mean was approximately 0.8138, with a standard deviation of 0.0619. The observed score range across all candidates was approximately 0.5597–0.8952.

Figure 2 compares the score distributions by candidate role. Faithful candidates had a higher mean score in this dataset, although the distributions overlap. A higher metric score does not by itself establish semantic correctness.

C. Pairwise Score Margins

For each pair, the score margin is defined as

$$\Delta_i = s_i^F - s_i^C, \tag{1}$$

where s_i^F is the intended faithful candidate score and s_i^C is the corrupted candidate score. Positive values indicate that the faithful candidate received a higher score, whereas negative values indicate ranking inversions.

The mean margin was 0.041884 and the median was 0.016785. The margins ranged from -0.006049 to 0.301254. Figure 3 displays the margin distributions by category.

Eighteen pairs had $|\Delta_i| < 0.005$, 37 had $|\Delta_i| < 0.010$, and 49 had $|\Delta_i| < 0.015$. These thresholds are exploratory and are not calibrated uncertainty estimates.

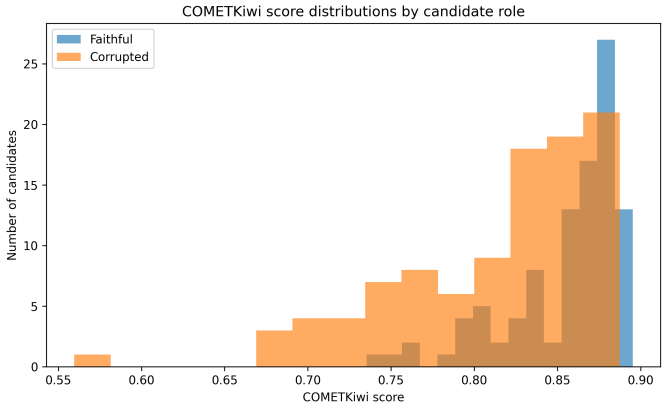


Fig. 2. COMETKiwi score distributions for intended faithful and corrupted candidates.

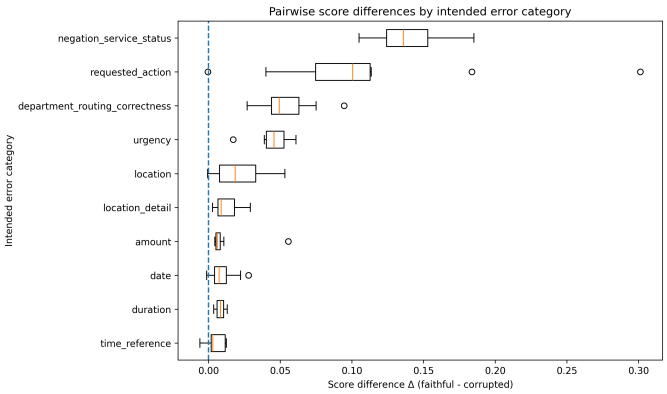


Fig. 3. Distribution of pairwise score differences by intended error category. The vertical dashed line marks zero; negative values indicate ranking inversions.

D. EDA Summary

The exploratory analysis shows that the challenge set contains ten pairs in each intended error category and that intended faithful candidates have a higher mean COMETKiwi score than corrupted candidates. However, some corrupted candidates receive higher scores than their paired faithful candidates. The score distributions overlap, and the score margin varies across pairs.

These observations motivate the pairwise and category-level analyses in Section VI. The dataset is synthetic, and its intended semantic labels have not been independently validated by bilingual human annotators. Therefore, this EDA describes the constructed dataset and does not establish performance on real-world citizen grievances.

V. Methodology

A. Challenge-Set Construction

The challenge set contains 100 synthetic English-source/Hindi-candidate pairs, identified as `SYN001_hi` through `SYN100_hi`. Ten intended error categories are represented, with ten pairs per category:

- negation and service status;

- date;
- duration;
- amount;
- location;
- location detail;
- requested action;
- department-routing correctness;
- time reference; and
- urgency.

For each source, the pair contains a candidate intended to preserve the source meaning and a candidate intentionally altered to introduce a targeted meaning change. The examples are synthetic and were designed for controlled diagnosis rather than sampled from real citizen submissions.

The category names describe the intended construction of the examples. In particular, department-routing examples contrast a wrong-department description with a correct-department description; they do not consistently swap the names of two specific departments.

B. Metric and Inference

We evaluated `Unbabel/wmt22-cometkiwi-da`. Each of the 200 Hindi candidates was scored against its English source using the reference-free metric. Inference was run on an NVIDIA RTX 3050 Laptop GPU with batch size 2.

The recorded SHA-256 digest of the evaluated dataset is:

`ae04ccf209516e531a8cd2badd31e7dece3d2f59ce1370f53b69adb51be6538d`

A positive Δ_i indicates that COMETKiwi ranks the intended faithful candidate higher; a negative value is a ranking inversion; and zero is a tie. The strict pairwise agreement rate is

$$A = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}(\Delta_i > 0), \tag{2}$$

where $N = 100$ and $\mathbb{I}(\cdot)$ is the indicator function. Ties are not counted as correct under this strict definition.

C. Analysis and Validity

We report the strict pairwise outcome, mean and median score differences, category-level outcomes, ranking inversions, and counts of low-margin comparisons. The low-margin thresholds are descriptive diagnostics rather than calibrated confidence boundaries.

The dataset labels reflect the intended synthetic construction. No independent bilingual human validation or human inter-annotator agreement study was performed. An AI-assisted semantic audit was used as a diagnostic aid, but it is not ground truth and does not replace human validation.

Since pairs share category-specific construction patterns, confidence intervals that assume independent observations may not fully reflect dependence. The experiment also does not measure the frequency of these errors in real-world data.

VI. Results

A. Overall Pairwise Ranking

COMETKiwi assigned a higher score to the intended faithful candidate in 95 of 100 pairs. In five pairs, the corrupted candidate received a higher score; no ties were observed. The strict pairwise agreement rate was therefore 95%.

The mean score difference was 0.041884, with a standard deviation of 0.053249. The median was 0.016785. Observed score differences ranged from -0.006049 to 0.301254.

TABLE I
Overall Pairwise Results

Measure	Result
Number of pairs	100
Faithful candidate ranked higher	95
Corrupted candidate ranked higher	5
Ties	0
Strict pairwise agreement	95.0%
Mean score difference	0.041884
Median score difference	0.016785
Standard deviation	0.053249
Minimum score difference	-0.006049
Maximum score difference	0.301254

A Wilson 95% confidence interval for the observed proportion 95/100 is approximately 88.82%–97.85%. This interval describes binomial sampling uncertainty under its assumptions; it does not account for the synthetic construction process or category-level dependence.

B. Category-Level Results

Table II summarizes pairwise outcomes and score differences by intended category. Each category contains ten pairs. COMETKiwi ranked the intended faithful candidate higher in all ten examples in six categories. Time reference had the lowest agreement, with eight correct rankings out of ten.

The time-reference category had both the lowest strict agreement and a small median margin. By contrast, the negation/service-status category had a comparatively large mean margin. These observations are descriptive and should not be generalized beyond the current synthetic examples.

C. Ranking Inversions

The five ranking inversions had negative score differences close to zero. Table III lists the affected pairs and their scores. These examples are useful for follow-up inspection, but they do not by themselves identify the mechanism behind a metric's ranking.

For all five inversions, the absolute score difference was below 0.01. This indicates that the ranking errors in this run were low-margin comparisons, although the scores alone cannot establish why the metric preferred the corrupted candidate. Potential explanations such as lexical overlap or contextual representation effects require separate controlled analysis and were not tested here.

TABLE II
PAIRWISE RESULTS BY INTENDED ERROR CATEGORY

Category	Pairs	Faithful higher	Mean Δ	Median Δ
Negation / service status	10	10	0.138982	0.135938
Department-routing correctness	10	10	0.055160	0.049298
Urgency	10	10	0.045549	0.045700
Location detail	10	10	0.012308	0.008710
Amount	10	10	0.011334	0.006096
Duration	10	10	0.008288	0.008297
Requested action	10	9	0.111335	0.100483
Location	10	9	0.021146	0.018569
Date	10	9	0.009803	0.007307
Time reference	10	8	0.004933	0.002971

TABLE III
PAIRS WHERE THE CORRUPTED CANDIDATE SCORED HIGHER

Pair ID	Category	Faithful	Corrupted	Intended contrast
SYN007_hi	Requested action	0.873128	0.873679	Restore a ration card versus cancel it
SYN025_hi	Location	0.885405	0.886033	Chandanpur versus Madanpur
SYN002_hi	Date	0.881748	0.883031	12 June versus 12 July
SYN079_hi	Time reference	0.877760	0.879352	Wednesday afternoon versus night
SYN059_hi	Time reference	0.735290	0.741339	Sunday afternoon versus night during a storm

D. Low-Margin Comparisons

Using $|\Delta_i| < 0.005$, 18 pairs were low-margin; using $|\Delta_i| < 0.010$, 37 pairs were low-margin; and using $|\Delta_i| < 0.015$, 49 pairs were low-margin.

At the 0.010 threshold, 32 pairs ranked the intended faithful candidate higher and five were ranking inversions. These thresholds are exploratory and should not be interpreted as model-calibrated uncertainty estimates.

VII. DISCUSSION

The main finding is that COMETKiwi ranked the intended faithful candidate higher for most pairs in this synthetic challenge set, while still producing five ranking inversions. The inversions involved requested action, location, date, and time reference. They illustrate why a high aggregate agreement rate does not imply that every potentially important administrative detail is reliably distinguished.

The category results caution against treating the score as a uniform guarantee across error types. Time-reference examples were the most difficult category in this run, whereas all ten examples in six other categories were ranked as intended. However, there are only ten examples per category, and categories are defined by the set's construction. These differences motivate larger, independently validated follow-up studies rather than strong claims about general category difficulty.

The experiment is not a deployment evaluation. It does not establish how frequently these errors occur in naturally submitted grievances, how a human operator would interpret the examples, or whether a metric score would change a public-service outcome. It also does not compare COMETKiwi with human judgments or other metrics.

VIII. LIMITATIONS AND ETHICAL CONSIDERATIONS

This pilot has several limitations. First, the 100 pairs are synthetic and may not capture the linguistic variety, spelling variation, code-switching, ambiguity, or contextual complexity of real citizen grievances. Second, the intended labels have not been independently verified by bilingual human annotators. The AI-assisted audit is only a diagnostic tool and cannot be presented as human validation or inter-annotator agreement.

Third, there are only ten pairs per category, limiting the reliability of category-level estimates. Fourth, all examples are evaluated with one metric checkpoint and one recorded inference setup; the study does not establish comparative performance against alternative evaluators. Fifth, the confidence interval for the overall proportion does not account for dependencies introduced by shared templates or category construction.

The examples concern potentially consequential administrative details, but the experiment measures metric rankings, not actual harms. The paper therefore does not infer observed harm or claim readiness for public-service deployment. Future work should use independently authored examples, bilingual expert review, adjudication of disputed labels, a larger sample, and real or carefully de-identified examples where ethically and legally appropriate. Any use in public services should retain human oversight and should not use a metric score as the sole basis for deciding how a grievance is handled.

IX. Reproducibility

The evaluated checkpoint was `Unbabel/wmt22-cometkiwi-da`; inference used an NVIDIA RTX 3050 Laptop GPU and batch size 2. The pairwise difference is defined in Eq. (1). The evaluated dataset version is identified by the SHA-256 digest reported in the methodology.

The recorded result artifacts include candidate scores, pairwise results, category analysis, failure cases, ranked-pair analysis, a summary JSON file, and a research-results note. Before release, the authors should publish the synthetic data, inference script, package versions, exact checkpoint revision, and generated result files in a stable repository, subject to the model's license conditions.

The COMETKiwi checkpoint is reported as distributed under a non-commercial share-alike license (CC-BY-NC-SA-4.0). Check its current license and comply with its terms before redistribution or commercial use.

X. Conclusion

This paper presented a controlled analysis of COMETKiwi on 100 synthetic English–Hindi citizen-grievance candidate pairs spanning ten intended administrative meaning-error categories. The metric ranked the intended faithful candidate higher in 95 pairs, with five low-margin ranking inversions and no ties. The time-reference category had the lowest pairwise agreement in this set.

The exploratory analysis showed that the faithful candidates had a higher mean score than corrupted candidates, while their score distributions overlapped. These findings provide a diagnostic snapshot of model behavior, not evidence of real-world harm or reliable performance on naturally occurring grievances. Independent bilingual validation, broader data, baseline comparisons, and human evaluation are necessary before drawing stronger conclusions about the use of quality-estimation metrics in public-service workflows.

References

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